Supplementary Material

for

PriEco-DRL: Coordinated Electric-Bus Eco-Driving under Transit-Priority Adaptive Signal Control via Deep Reinforcement Learning

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This supplementary document provides additional experimental details and extended results that support the findings reported in the main manuscript. In particular, it includes additional trajectory-level comparisons, trip-level energy profiles, and representative bus longitudinal profiles that could not be fully presented in the main text due to space limitations.

S1 Behavioral evidence for the A2 ablation

To further examine the behavior underlying the large energy–delay trade-off of PriEco-DRL-B-A2, Fig. S1 compares its EB speed and absolute-acceleration distributions with those of the complete PriEco-DRL-B model. All one-second records while an EB remains in the simulation network are included, covering regular driving, scheduled dwell, signal waiting, queueing, and vehicle-entry states.

A2 exhibits a pronounced concentration around approximately 2.3 m/s and substantially less medium- and high-speed operation, while its absolute-acceleration distribution is more concentrated near zero. Importantly, the fraction of in-network time with $v \leq 0.1$ m/s is lower for A2 than for PriEco-DRL-B (20.99% versus 25.30%). Scheduled dwell accounts for only 1.60% and 2.71% of the in-network time, respectively. The large delay increase of A2 is therefore associated mainly with prolonged very-low-speed progression rather than a higher fraction of complete stops.

S2 Additional Trajectory-Level Analyses

This section provides additional trajectory-level evidence that complements the line-level results in the main paper. In particular, we present representative space–time trajectories, trip-level energy profiles, and detailed longitudinal speed/acceleration/jerk comparisons for selected EBs from unseen lines. These results further illustrate how PriEco-DRL reduces non-scheduled stop–start events, smooths speed transitions, and selects different operating points along the energy–time trade-off.

S2.1 Representative space–time trajectories

Fig. S2 presents representative longitudinal space–time trajectories of electric buses from four unseen lines under the selected control methods. Panels (a) and (b) correspond to the beginning and the middle of the 3-hour simulation, respectively. Because adaptive signal controllers

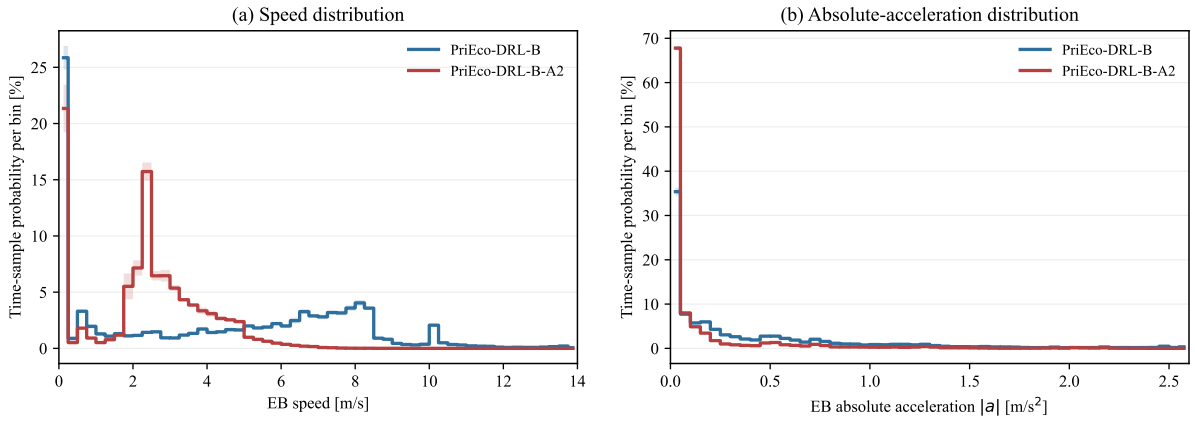


Figure S1: Behavioral distributions of PriEco-DRL-B and the A2 ablation: (a) EB speed and (b) absolute longitudinal acceleration. All one-second records while an EB remains in the simulation network are included, covering regular driving, scheduled dwell, signal waiting, queueing, and vehicle-entry states. The distributions are normalized over the evaluation samples. A2 exhibits a pronounced concentration around very low operating speeds and near-zero acceleration, indicating prolonged low-speed progression rather than simply more complete stopping.

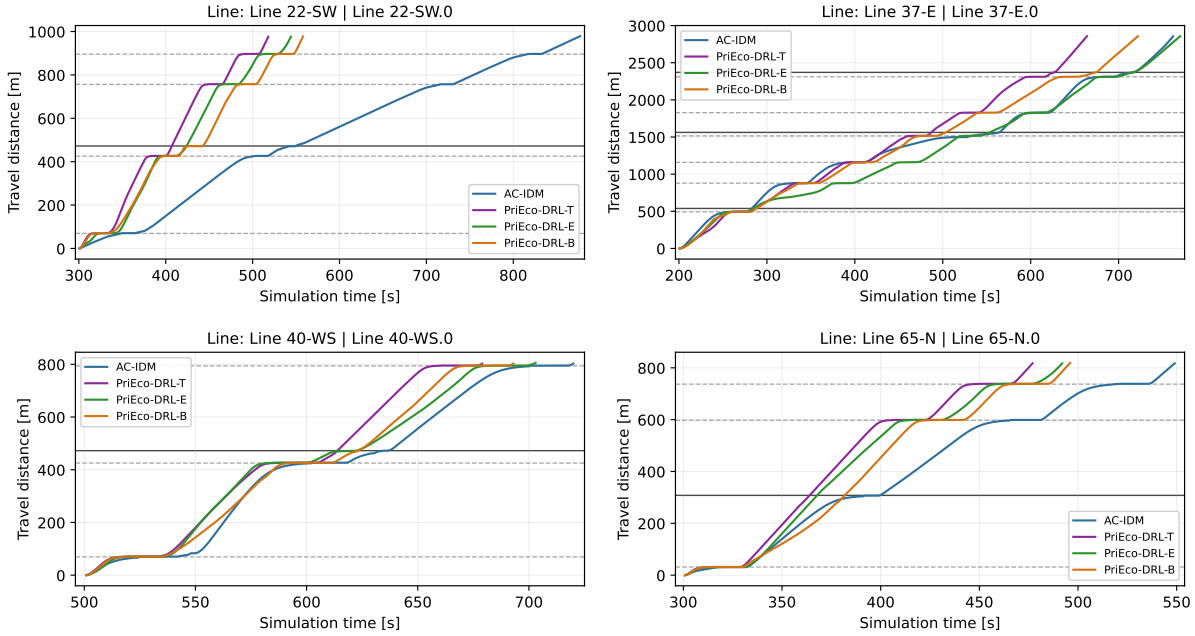
generate method-specific phase sequences, the exact switching instants are not directly comparable across methods; therefore, only intersection locations (solid horizontal lines) and bus stops (dashed lines) are marked.

During the low-demand period (Fig. S2a), AC-IDM exhibits clear progression slowdowns around intersections, with longer flat segments near the intersection markers indicating non-scheduled waiting. In contrast, all PriEco-DRL variants achieve faster progression and shorter non-scheduled holds, which is consistent with the line-level reduction in travel time reported in the main text. The difference is especially visible on longer trajectories such as Line 37-E.0, where PriEco-DRL maintains more continuous advancement over successive intersections. Among the variants, PriEco-DRL-T generally exhibits the steepest effective slope, reflecting a stronger emphasis on travel-time efficiency, whereas PriEco-DRL-E adopts a more conservative progression pattern. PriEco-DRL-B typically lies between the two.

In the congested period (Fig. S2b), all methods experience slower progression due to longer queues and intermittent spillbacks, and some buses (e.g., Line 22-SW.9) undergo extended waiting at intersections. Under such conditions, uninterrupted green access becomes difficult for all methods, so the main advantage of PriEco-DRL comes from bus-side pacing rather than guaranteed non-stop passage. Even in these cases, PriEco-DRL reduces large stop–start disruptions and yields a more regular advancement pattern. For example, for Line 37-E.6 and bus_40Mui.4, the PriEco-DRL trajectories progress more steadily across successive intersections and show shorter non-scheduled holds than AC-IDM. Overall, these trajectory-level observations support the two main mechanisms identified in the paper: (i) reducing intersection-induced stop–start events through anticipative approach-speed regulation, and (ii) shaping smoother speed profiles that avoid unnecessary high-power transients, with the specific pacing determined by the reward-weight setting.

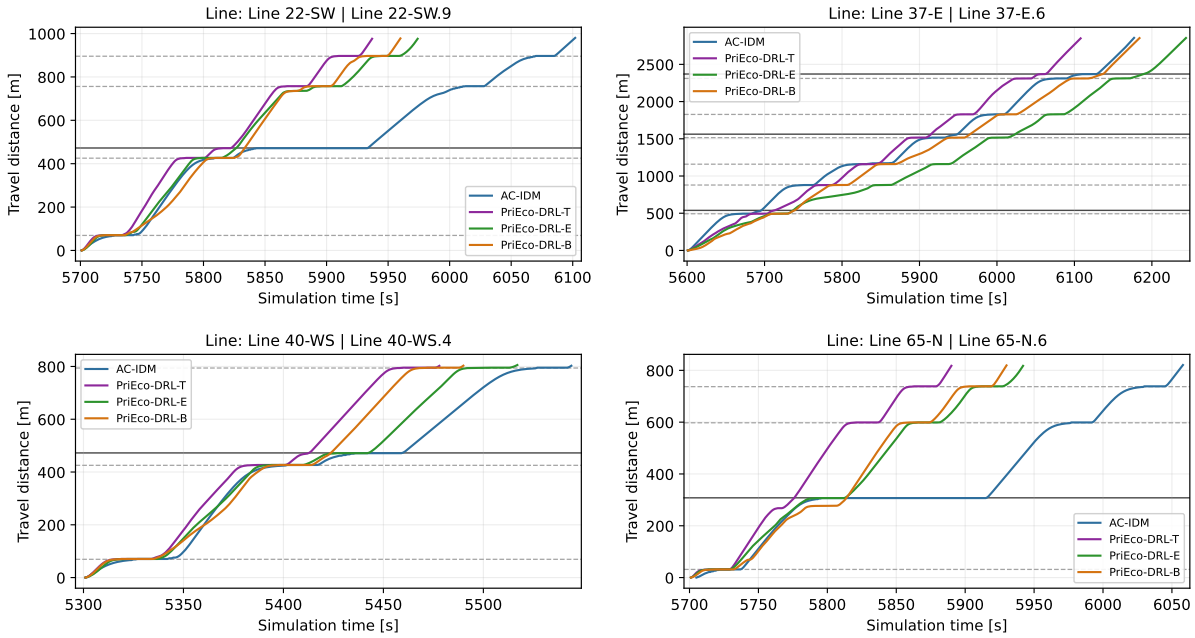
S2.2 Additional trip-level energy profiles

Fig. S3 further illustrates the trip-level energy-consumption dynamics by plotting (i) the cumulative electrical energy consumption (solid curves, left axis) and (ii) the step-wise net energy increment (shaded bars, right axis) for four representative EBs from unseen lines during the congested period. The shaded series reveals how energy accumulates under mixed traffic and



(a) EB trajectories of the unseen four lines at the start of the simulation with light traffic volume

(a) EB trajectories of the unseen four lines at the start of the simulation with light traffic volume



(b) EB trajectories of the unseen four lines in the middle of the simulation with heavy traffic volume

(b) EB trajectories of the unseen four lines in the middle of the simulation with heavy traffic volume

Figure S2: Space–time trajectories on four unseen lines during (a) light and (b) congested traffic. PriEco-DRL uses evaluation seed 5; AC-IDM traces are digitized from the original supplementary figure. Solid and dashed horizontal lines mark intersections and stops, respectively; the horizontal axis is absolute simulation time.

adaptive SPaT: large positive spikes typically correspond to high-power traction events such as re-acceleration after queuing or stop–go recovery, whereas smaller increments indicate smoother progression with fewer abrupt power demands. Negative increments, when present, correspond to regenerative braking, i.e., partial recovery of kinetic or potential energy during deceleration.

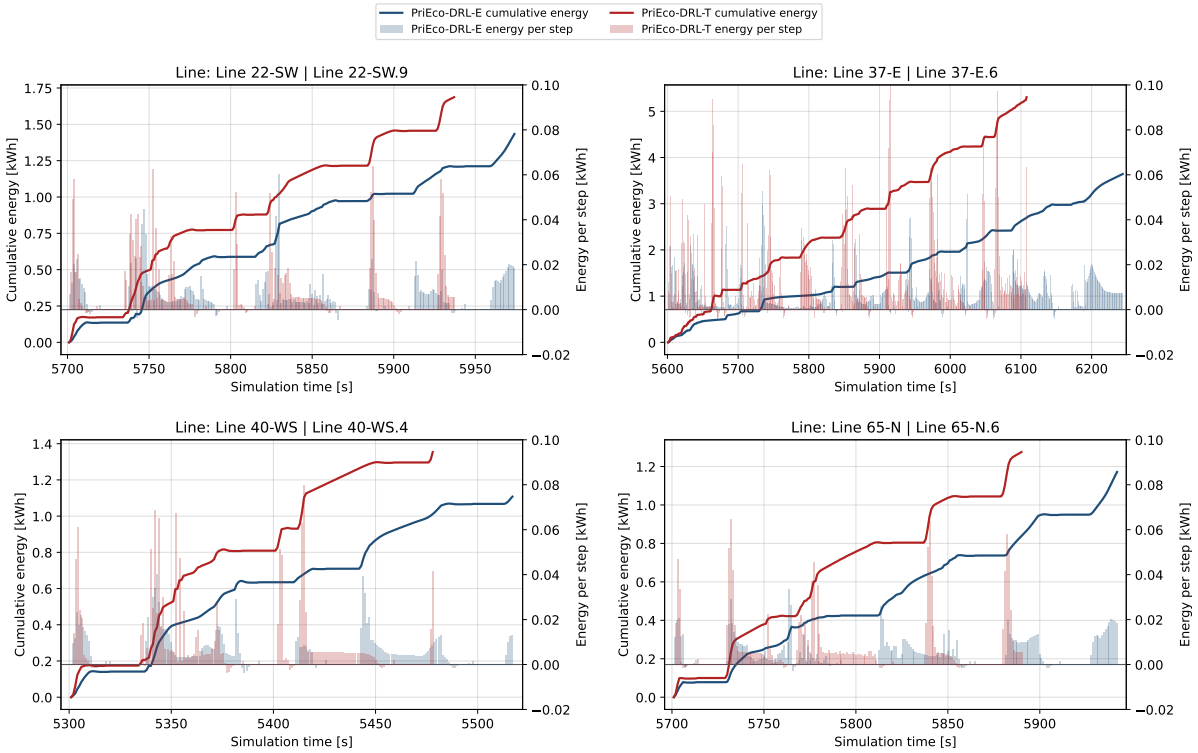


Figure S3: Energy profiles of four representative unseen-line trips under PriEco-DRL-T/E during congestion (evaluation seed 5). Solid curves show cumulative electrical energy (left axis); bars show signed per-step energy increments (right axis). Negative increments indicate regenerative energy recovery.

Comparing the two reward-weight variants, PriEco-DRL-E consistently yields lower cumulative energy trajectories than PriEco-DRL-T across all four trips, and the gap grows as the trip proceeds. This difference can be traced to the step-wise profiles: PriEco-DRL-T exhibits more frequent and larger energy spikes (e.g., for Line 37-E.6 and Line 22-SW.9), indicating repeated high-power transients associated with aggressive progression and stop–go recovery in congestion. In contrast, PriEco-DRL-E suppresses both the frequency and magnitude of these spikes, suggesting that stronger energy-oriented reward weighting encourages more moderate approach and discharge speeds, fewer unnecessary stop–start maneuvers, and a lower-variance speed profile. These effects accumulate over time and explain the lower total energy use of PriEco-DRL-E observed in the line-level results.

S2.3 Representative EB longitudinal profiles

To further explain the time–energy trade-off among the PriEco-DRL variants, Fig. S4 compares the longitudinal profiles of a representative EB trip (Line 22-SW.9) during the congested period. Panels (a)–(c) show the speed, acceleration, and jerk trajectories, and panel (d) reports cumulative energy consumption.

The time-prioritized variant PriEco-DRL-T pursues more aggressive progression, reaching higher cruising speeds and exhibiting more pronounced acceleration events. In contrast, the energy-prioritized variant PriEco-DRL-E adopts a smoother and more conservative pacing strategy, with fewer high-speed excursions and milder acceleration/deceleration swings. PriEco-DRL-B provides an intermediate behavior, combining relatively steady progression with moderated actuation, although localized transients can still appear depending on the surrounding traffic and signal interactions.

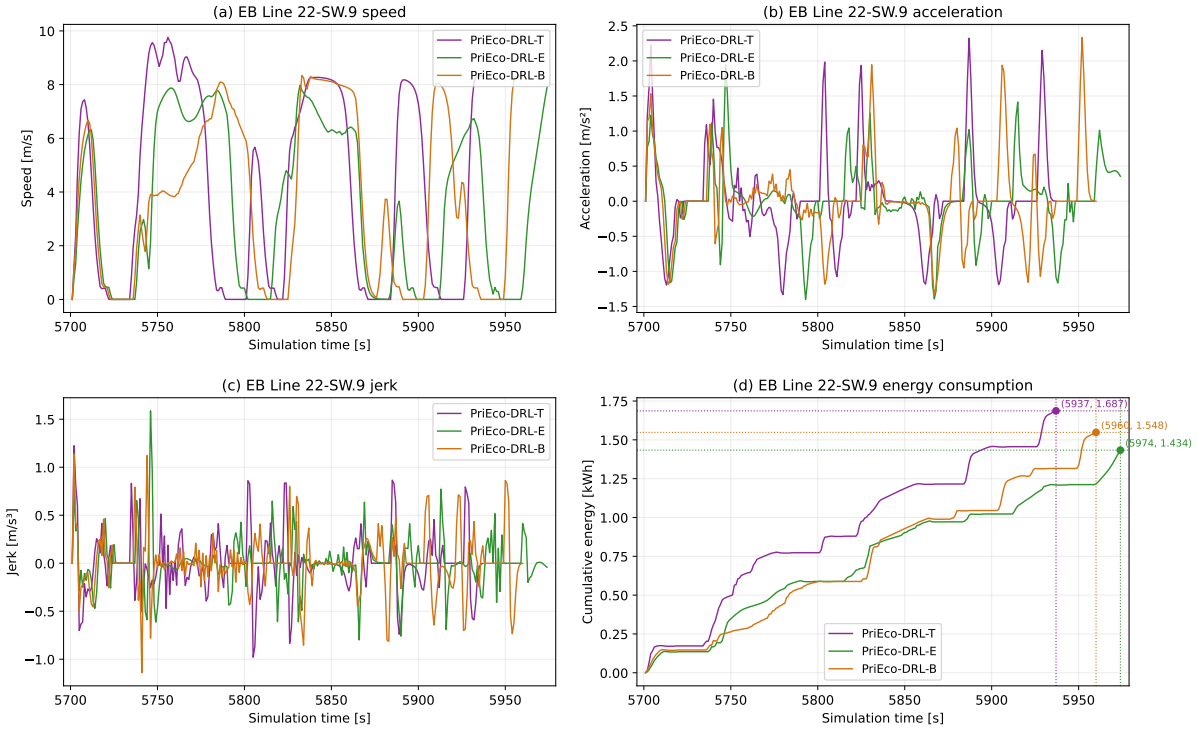


Figure S4: Longitudinal profiles of the congested-period trip Line 22-SW.9 under PriEco-DRL-T/E/B (evaluation seed 5): (a) speed, (b) acceleration, (c) jerk, and (d) cumulative electrical energy.

For this representative trip, PriEco-DRL-T completes the trip first with the highest energy use, PriEco-DRL-E achieves the lowest energy use with a longer travel time, and PriEco-DRL-B lies between them.

S3 Trajectory-based safety diagnostics across evaluation seeds

One-second logs from the 15 evaluations record zero collision-positive samples. T and B have zero samples below the 1.5 s TTC reference; E has one, corresponding to a mean seed-specific exposure of 0.0006%. E also has the lowest strong-deceleration event rate (Table S1). Each variant uses one fixed policy evaluated with traffic seeds 1–5, and each run contains 102 buses.

Evaluation TTC uses bumper-to-bumper clearance divided by closing speed $\Delta v_t = v_t - v_{\mathrm{lead},t}$ when a leader is present and $\Delta v_t > 10^{-3}$ m/s; otherwise TTC is infinite. The archived quantity $\mathrm{TTC}_t^{\mathrm{rec}}$ uses the SUMO `getLeader` gap, which subtracts $g_{\min} = 2.5$ m from bumper clearance. All 1,530 saved bus trips use the unchanged vehicle type with this minimum-gap setting. For forward traffic, $0 < \Delta v_t \le v_t$, giving

$$\mathrm{TTC}_t^{\mathrm{B}} = \mathrm{TTC}_t^{\mathrm{rec}} + \frac{g_{\min}}{\Delta v_t},$$
$$L_t = \mathrm{TTC}_t^{\mathrm{rec}} + \frac{g_{\min}}{v_t} \;\le\; \mathrm{TTC}_t^{\mathrm{B}}.$$

The bound L_t certifies the 1.5 s threshold wherever $L_t \ge 1.5$ s. Remaining threshold statuses are determined through logging-only, same-seed replays, whose original bus-trajectory and network-time-series prefixes are reproduced exactly. The replay covers T-seed5 to 5,636 s, E-seed2 to 7,663 s, and E-seed5 to 5,213 s. All 479,223 original records are retained. Thus the 1.5 s exposure in Table S1 is fully classified. For 3.5 s exposure, measured TTC is used at replayed records and L_t elsewhere, giving an upper bound. A measured minimum TTC is reported

Table S1: Network safety diagnostics across five seeds per variant using bumper-clearance TTC.

Method	Collision-positive records	TTC < 1.5 s [%]	TTC < 3.5 s upper bound [%]	Min. TTC [s]	Strong deceleration [events/100 bus-km]
PriEco-DRL-T	0	0	≤ 0.6470	≥ 1.5531	35.52 ± 7.65
PriEco-DRL-E	0	0.0006	≤ 0.4710	1.3344	14.69 ± 2.02
PriEco-DRL-B	0	0	≤ 0.4190	≥ 1.6102	42.07 ± 7.44

All original records are retained: 147,420, 169,458, and 162,345 for T/E/B. Percentages average the five seed-specific values. The 1.5 s column is fully classified; $\leq$ and $\geq$ denote conservative bounds.

when it is certified against every remaining lower bound; otherwise a lower bound is marked $\geq$. Bounds are rounded outward, and percentages are averaged equally over the five seeds. The training-reward TTC retains its 1 m gap margin and 0.1 m/s closing-speed floor.

Braking diagnostics use the logged one-second speed-difference acceleration, $\bar{a}_t = (v_t - v_{t-1})/\Delta t$, with $\Delta t = 1$ s. A literature-referenced harsh-braking criterion is $\bar{a}_t < -3$ m/s^2 [1]. All 15 runs record zero exceedance episodes at this criterion; their minimum logged acceleration is -2.6 m/s^2, consistent with the bounded action interface. To characterize braking intensity within this range, Table S1 reports strong-deceleration episodes at $\bar{a}_t < -2$ m/s^2, using the braking penalty's lower reference value. Consecutive qualifying samples per bus are merged into one recorded episode. Distance-normalized rates are reported as mean $\pm$ sample standard deviation across traffic seeds.

The diagnostics characterize collision, following, and braking behavior at the recorded one-second resolution.

S3.1 Targeted cut-in and sudden lead-braking tests

The frozen T/E/B policies undergo 24 distinct challenges each, giving 72 tests with the original network, observation builder, normalization statistics, acceleration interface, and EB vehicle settings. Each test contains one EB and one scripted car and records 30 s after insertion, with 1 s control and sampling intervals. Cut-in tests combine initial EB speeds of 5.00, 6.67, and 8.33 m/s, closing speeds of 2 and 4 m/s, and initial longitudinal bumper clearances of 20 and 30 m. The car changes from lane 2 to the EB's lane 1 on the three-lane approach and continues onto the downstream road. The movement is held green to isolate the following response; these speeds respect the downstream limit.

Lead-braking tests use a longer road with a 13.89 m/s limit, equal initial vehicle speeds of 8.33, 11.11, and 13.89 m/s, and car decelerations of 3.5 or 5 m/s^2 until stationary. Initial bumper clearance is set by a one-control-period stopping-distance diagnostic,

$$g_0 = s_{\text{reac}} + \frac{v_+^2}{2b_{\text{EB}}} - \frac{v_0^2}{2b_{\text{lead}}} + g_{\min} + m, \tag{S1}$$

where $v_+ = \min(v_0 + a_{\max}\Delta t, v_{\max})$ and $s_{\text{reac}} = \int_0^{\Delta t} \min(v_0 + a_{\max}u, v_{\max})\, du$ allow for continued acceleration during the first control interval. We use $b_{\text{EB}} = a_{\max} = 2.6$ m/s^2, $v_{\max} = 13.9$ m/s, $g_{\min} = 2.5$ m, and margins $m \in \{5, 15\}$ m. The complete grid is fixed before simulation and uses seed 1; variation comes from the distinct initial conditions.

All 72 tests complete with positive bumper clearance and zero recorded collisions or teleports (Table S2). The scripted car's full speed profile and actual lane change are verified from the event records. The EB retains its original `speedMode=31`, including vehicle-following checks. Both requested and realized acceleration are recorded: for example, T requests positive acceleration throughout its cut-in tests, while the execution model also produces deceleration. The outcomes therefore characterize each frozen policy together with its original execution model.

Table S2 reports directly computed bumper-clearance TTC from the saved leader gaps and relative speeds. The minimum is 1.702 s across all 72 tests, with zero samples below 1.5 s.

Table S2: Frozen-policy cut-in and lead-braking tests using bumper-clearance TTC.

Setting	Challenge	Cases	Collision cases	Min. TTC [s]	TTC < 1.5 s [cases]	Min. bumper gap [m]
T	Cut-in	12	0	2.114	0	4.000
T	Lead braking	12	0	2.415	0	2.500
E	Cut-in	12	0	3.094	0	4.000
E	Lead braking	12	0	2.386	0	2.500
B	Cut-in	12	0	2.973	0	4.000
B	Lead braking	12	0	1.702	0	2.500

These measurements complement the archived-network bounds and characterize the specified challenges at 1 s resolution.

S4 Reward scaling and fixed-auxiliary-weight sensitivity

The reward definitions in Section V-B use component-specific reference scales and bounds. Energy is scaled by $E_{\max}$ and squared jerk by J_0; progress, stop-approach, and signal-guidance terms use clipping or bounded mappings, while event penalties retain their specified coefficients. The outer weights express numerical reward emphasis, and physical energy and delay quantify the resulting operational outcomes.

We examine 35 archived evaluations covering 17 energy–efficiency settings of the form $(\omega_e, \omega_t, \omega_o) = (2(1-w), 2w, 1)$, where $w = \omega_t/(\omega_e + \omega_t)$ ranges from 0 to 0.95. The auxiliary coefficient remains fixed to retain a common scaling of auxiliary comfort, safety, and operational penalties while examining energy–efficiency preferences. The archived implementation supplies the normalized coefficient vector to a preference-conditioned policy; this vector is held fixed throughout each evaluation, giving fixed scalarization coefficients within each run. The main T/E/B results come from separately trained fixed-weight policies and retain their original evaluation protocol.

Figure S5 summarizes the 35 evaluations as 17 configuration means in the energy–delay plane. Each setting averages its 1–4 available evaluations. The line joins the 15 empirically non-dominated means among these settings as a visual guide, and the other two means remain visible. Greater time-efficiency emphasis generally corresponds to higher energy use and lower EB delay. This fixed-auxiliary slice illustrates the observed energy–efficiency trade-off.

References

[1] M. Dodd, I. Knight, A. Edwards, M. Edwards, P. Ball, J. Appleby, and J. Radcliffe, "Bus safety standard: Bus braking data analysis," TRL Limited, Tech. Rep. PPR1013, Feb. 2022, prepared for Transport for London; see p. 28. [Online]. Available: https://www.trl.co.uk/uploads/trl/documents/PPR1013---Bus-Safety-Standard---Bus-Braking-Data-Analysis_1.pdf

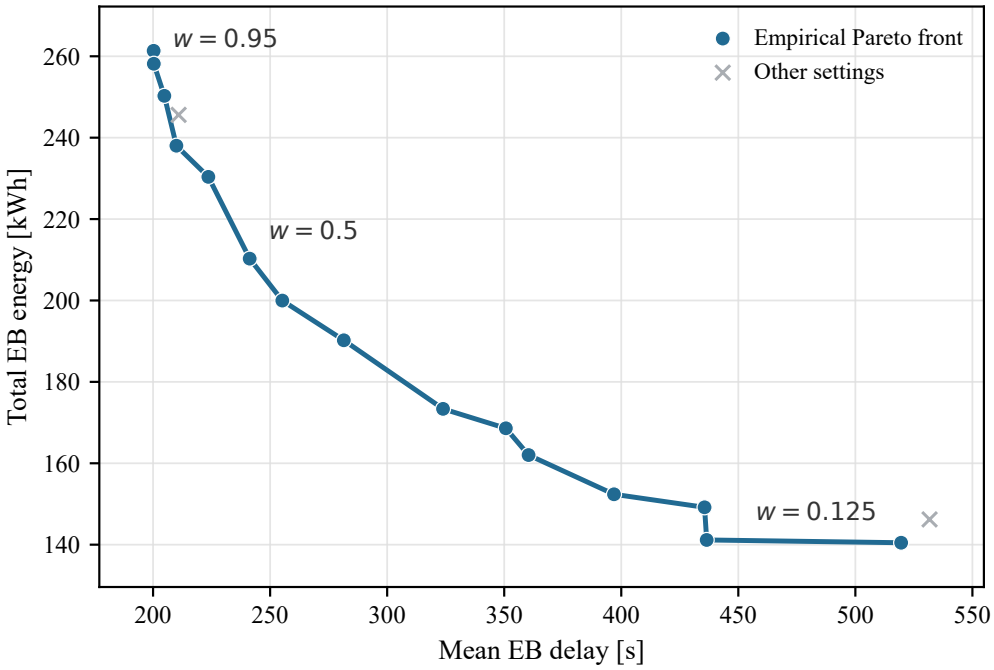


Figure S5: Energy–delay trade-off with $\omega_o = 1$. Each marker represents a configuration mean; the line joins empirically non-dominated means among the evaluated settings.